

MEGAN TJANDRASUWITA

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EDUCATION

Massachusetts Institute of Technology Ph.D. in Computer Science NSF Graduate Research Fellowship Program (GRFP)	2022 - Present
Massachusetts Institute of Technology M.S. in Computer Science, GPA 5.0/5.0 MIT Presidential Fellowship, NSF Graduate Research Fellowship Program (GRFP)	2022 - 2024
California Institute of Technology B.S. in Computer Science, GPA: 4.0/4.0	2018 - 2022

RESEARCH INTERESTS

Multimodal AI for human behavior analysis and social reasoning. Representation similarity in multimodal AI. Machine learning methods that combine deep learning with symbolic knowledge.

PUBLICATIONS

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- **Megan Tjandrasuwita**, Jie Xu, Armando Solar-Lezama, Wojciech Matusik. **MeMo: Meaningful, Modular Controllers via Noise Injection.** *NeurIPS 2024*. [[paper](#)]
 - Liane Makatura, Michael Foshey, ..., **Megan Tjandrasuwita** (7th author), ... **How Can Large Language Models Help Humans in Design and Manufacturing?** *Harvard Data Science Review 2024*. [[part1](#)], [[part2](#)]
 - Tailin Wu, **Megan Tjandrasuwita**, Zhengxuan Wu, Xuelin Yang, Kevin Liu, Rok Sasic, Jure Leskovec. **ZeroC: A Neuro-Symbolic Model for Zero-shot Concept Recognition and Acquisition at Inference Time.** *NeurIPS 2022, ICML 2022 Beyond Bayes Workshop*. [[paper](#)]
 - Jennifer J. Sun*, **Megan Tjandrasuwita***, Atharva Sehgal*, Armando Solar-Lezama, Swarat Chaudhuri, Yisong Yue, Omar Costilla-Reyes. **Neurosymbolic Programming for Science.** *NeurIPS 2022 AI4Science Workshop*. [[paper](#)]
 - **Megan Tjandrasuwita**, Jennifer J. Sun, Ann Kennedy, Swarat Chaudhuri, Yisong Yue. **Interpreting Expert Annotation Differences in Animal Behavior.** *CVPR 2021 CV4Animals Workshop*. [[paper](#)]
 - Jasmina Arifovic, Anil Donmez, John Ledyard, **Megan Tjandrasuwita**. **Individual Evolutionary Learning and Zero-Intelligence in the Continuous Double Auction.** *Handbook of Experimental Finance, Chapter 19, p.225 – p.250, Edward Elgar publishing*

RESEARCH EXPERIENCE

Multisensory Intelligence Lab <i>Deep Learning Researcher</i>	Oct 2024 - Present <i>Advisor: Professor Paul Liang</i>
• Develop a machine learning framework to train neural networks on modalities with varying degrees of comparability and to measure representation similarity. • Comprehensive evaluation across MLP and Transformer models on supervised and unsupervised tasks involving text, vision, and audio.	
The Fiete Lab <i>Computational Neuroscience Researcher</i>	Apr 2024 - Oct 2024 <i>Advisor: Professor Ila Fiete</i>

- Develop a computational model for mapping and navigating family trees based on the Entorhinal-Hippocampal memory circuit, enabling reasoning over learned structures.

Computer-Aided Programming Group

Sep 2022 - Present

Deep Learning Researcher

Advisor: Professor Armando Solar-Lezama

- Researched learning reusable, modular controllers for robots using a novel training objective.
- Developed ML pipeline involving imitation learning for pretraining neural network modules and reinforcement learning for transferring modules to different structures and tasks.
- Experiments in locomotion and grasping domains demonstrate that the learned modules outperform graph neural network and Transformer baselines in training efficiency.
- Full-conference paper published at *NeurIPS 2024*.

Stanford Network Analysis Project, Stanford

Jun 2021 - Jun 2022

Deep Learning Researcher

Advisor: Professor Jure Leskovec

- Researched neurosymbolic methods for performing human-like concept recognition and reasoning in visual domains.
- Constructed policy architecture of agent that performs unsupervised object discovery and identifies relevant relations between concepts. Integrated recurrence in graph neural network (GNN) architecture to improve agent's long-term reasoning.
- Full-conference paper published at *NeurIPS 2022*. Short paper published at *ICML 2022 Beyond Bayes Workshop*.

Machine Learning Group, Caltech

Sep 2020 - Jun 2022

Machine Learning Researcher

Advisor: Professor Yisong Yue

- Employed program synthesis to generate human-interpretable programs that classify animal behavior based on laboratory datasets.
- Resulting programs exceeded accuracy of baseline classifiers, and visualizations of temporal filters were more interpretable to neuroscientists.
- First-authored paper published at the *CVPR 2021 CV4Animals Workshop*.

INDUSTRY EXPERIENCE

Oracle - Corporate Architecture

Jun 2020 - Sep 2020

Software Engineer Intern

- Applied machine learning algorithms to create a recommendation system that processes purchase requests from Oracle's employees.
- Achieved highly interpretable outputs through its ranking system and had a precision of over 80% on validation examples.

TEACHING EXPERIENCE

Caltech Teaching Assistant

Jan 2022 - Jun 2022

- Held office hours, graded problem sets, and prepared lecture notes for graduate-level computer science courses: "Machine Learning and Data Mining" (CS/EE 155) and "Advanced Machine Learning Methods" (CS/EE 159).

HONORS & AWARDS

NSF Graduate Research Fellowship (GRFP)	2022
MIT Stata Family Presidential Fellowship	2022